Advanced Methods in Deep Learning

Lecture 2: Non-linearities, Loss Functions, Backpropagation & Gradient Descent

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Today's Roadmap

Recap and Activation Functions Deep Dive

Loss Functions in Detail

The Chain Rule

Backpropagation Algorithm

Gradient Descent Variants

PyTorch Implementation

Summary and Next Steps

Recap & Activation Functions

Recap from Lecture 1

What we learned:

- A neural network computes: $\mathbf{h}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)})$.
- The **forward pass** maps input $\mathbf{x}$ to prediction $\hat{y}$.
- We need a **loss function** $\mathcal{L}$ to measure how bad our prediction is.
- We use **gradient descent** to update parameters: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} \mathcal{L}$.

Today's question:

How do we compute $\nabla_{\mathbf{w}} \mathcal{L}$ efficiently for a deep network?

Answer: The **backpropagation** algorithm — a systematic application of the chain rule.

Activation Functions: Derivatives Matter

To do backpropagation, we need the **derivative** of each activation function.

Name	$\sigma(z)$	$\sigma'(z)$	**Range**
Sigmoid	$\frac{1}{1+e^{-z}}$	$\sigma(z)(1-\sigma(z))$	$(0,1)$
Tanh	$\frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$	$1-\tanh^2(z)$	$(-1,1)$
ReLU	$\max(0,z)$	1 if $z>0$; 0 if $z<0$	$[0,\infty)$
Leaky ReLU	$\max(\alpha z,z)$	1 if $z>0$; α if $z<0$	$(-\infty,\infty)$

Deriving the Sigmoid Derivative

Let $\sigma(z) = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1}$. Apply the chain rule:

Step 1: $\sigma'(z) = -(1+e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}$

Step 2: Rewrite in terms of $\sigma(z)$:

$$\sigma'(z) = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z) \cdot \frac{(1+e^{-z})-1}{1+e^{-z}} = \sigma(z)(1-\sigma(z))$$

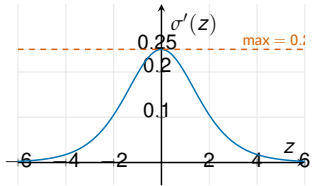
Result

$$\boxed{\sigma'(z) = \sigma(z)\big(1-\sigma(z)\big)}$$

Note: Max value $\sigma'(0) = 0.25$. This causes the vanishing gradient problem.

The Vanishing Gradient Problem

Sigmoid derivative peaks at 0.25:



Problem for deep networks:
With L layers of sigmoid, the gradient at layer 1 involves:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} \propto \prod_{l=1}^{L} \sigma'(z^{(l)})$$

Since $\sigma'(z) \leq 0.25$:

$$\prod_{l=1}^{L} 0.25 = 0.25^{L}$$

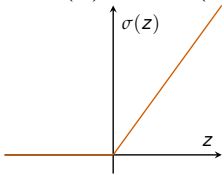
For $L = 10$: $0.25^{10} \approx 10^{-6}$

Gradients vanish exponentially with depth!

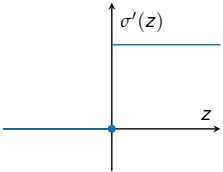
Solution: Use **ReLU**, where $\sigma'(z) = 1$ for $z > 0$. No vanishing gradient!

Why ReLU Dominates

ReLU: $\sigma(z) = \max(0, z)$



ReLU derivative:



- Derivative $= 1$ for $z > 0$: no vanishing gradient.
- Computationally cheap.
- Sparse activation (many neurons output 0).

Problem: "Dead ReLU" — if $z < 0$ always, the neuron never updates.

Fix: Leaky ReLU ($\alpha = 0.01$), ELU, GELU.

Softmax: From Binary to Multi-class Classification

For K classes, we need a probability distribution over classes.

Softmax Function

Given logits $z_1, z_2, \ldots, z_K$ (raw outputs of the network), the **softmax** function computes:

$$p(y = k|\mathbf{x}) = \text{softmax}(z_k) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}} \qquad \text{for } k = 1, \ldots, K$$

Properties:

- $\sum_{k=1}^{K} p(y = k|\mathbf{x}) = 1$ (valid probability distribution).
- $p(y = k|\mathbf{x}) > 0$ for all k (always positive).
- Larger $z_k \Rightarrow$ higher probability for class k.
- When $K = 2$: softmax reduces to the sigmoid function.

Softmax: Numerical Example

Network output (logits) for $K = 3$ classes:

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 2.0 \\ 1.0 \\ 0.1 \end{pmatrix}$$

Step 1: Compute exponentials:

$$e^{z_1} = e^{2.0} = 7.389, \quad e^{z_2} = e^{1.0} = 2.718, \quad e^{z_3} = e^{0.1} = 1.105$$

Step 2: Sum of exponentials:

$$\sum_{j=1}^{3} e^{z_j} = 7.389 + 2.718 + 1.105 = 11.212$$

Step 3: Normalize:

$$p_1 = \frac{7.389}{11.212} = 0.659, \quad p_2 = \frac{2.718}{11.212} = 0.242, \quad p_3 = \frac{1.105}{11.212} = 0.099$$

Prediction: class 1 with probability 65.9%. Check: $0.659 + 0.242 + 0.099 = 1.0$ ✓

Loss Functions in Detail

Loss Functions: Overview

The loss function $\mathcal{L}$ measures how far our prediction $\hat{y}$ is from the truth y.

Task	**Loss**	**Formula**	**Output activation**
Regression	MSE	$\frac{1}{N}\sum(y_i - \hat{y}_i)^2$	None (linear)
Binary classif.	BCE	$-\frac{1}{N}\sum[y \log \hat{y} + (1-y)\log(1-\hat{y})]$	Sigmoid
Multi-class	CE	$-\frac{1}{N}\sum\sum y_{ik} \log \hat{y}_{ik}$	Softmax

Key principle: the choice of loss function depends on the task. The loss determines what the network optimizes for.

MSE Loss: Gradient Derivation

Loss for one sample: $\ell = (y - \hat{y})^2$ where $\hat{y} = f(\mathbf{x}; \theta)$.

Gradient with respect to the prediction:

$$\frac{\partial \ell}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}}(y - \hat{y})^2 = 2(y - \hat{y}) \cdot (-1) = -2(y - \hat{y})$$

Average over N samples:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}_i} = -\frac{2}{N}(y_i - \hat{y}_i)$$

Interpretation:

- The gradient is proportional to the **error** $(y_i - \hat{y}_i)$.
- Large errors produce large gradients $\Rightarrow$ faster learning.
- This is the **starting point** of backpropagation: we first compute $\frac{\partial \mathcal{L}}{\partial \hat{y}}$, then propagate backward.

Binary Cross-Entropy: Derivation from MLE

Assume $y \in \{0, 1\}$ and model predicts $\hat{y} = p(y = 1|\mathbf{x})$. Likelihood of one sample: $p(y|\mathbf{x}) = \hat{y}^y (1 - \hat{y})^{1-y}$.

Log-likelihood $\rightarrow$ **Negative log-likelihood** (= BCE):

$$\ell = -\big[y \log \hat{y} + (1 - y) \log(1 - \hat{y})\big]$$

Gradient w.r.t. $\hat{y}$**:**

$$\frac{\partial \ell}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

Gradient w.r.t. logit z (where $\hat{y} = \sigma(z)$, $\sigma' = \sigma(1 - \sigma)$):

$$\frac{\partial \ell}{\partial z} = \frac{\partial \ell}{\partial \hat{y}} \cdot \sigma'(z) = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})} \cdot \hat{y}(1 - \hat{y}) = \hat{y} - y$$

The Beautiful Cancellation: BCE + Sigmoid

Key Result

$$\frac{\partial \ell}{\partial z} = \sigma(z) - y = \hat{y} - y$$

The gradient of BCE w.r.t. the logit z is simply $\hat{y} - y$.

Why is this beautiful?

- No $\sigma'(z)$ term in the gradient!
- Even when sigmoid saturates, gradient is large if prediction is wrong.
- **Not a coincidence:** BCE was designed to pair with sigmoid.
- Compare: MSE + sigmoid $\Rightarrow \frac{\partial \ell}{\partial z} = -2(y - \hat{y})\sigma'(z)$ – vanishes!

Lesson: Always pair the right loss with the right activation.

Categorical Cross-Entropy Loss

For K-class classification with one-hot encoding $\mathbf{y} = (y_1, \ldots, y_K)$:

Categorical Cross-Entropy

$$\mathcal{L} = -\sum_{k=1}^{K} y_k \log \hat{y}_k = -\log \hat{y}_c$$

where c is the true class (since only $y_c = 1$, all other $y_k = 0$).

Example: True class $= 2$ (i.e., $\mathbf{y} = (0, 1, 0)$), predictions $\hat{\mathbf{y}} = (0.1, 0.7, 0.2)$:

$$\mathcal{L} = -(0 \cdot \log 0.1 + 1 \cdot \log 0.7 + 0 \cdot \log 0.2) = -\log(0.7) = 0.357$$

If predictions were $\hat{\mathbf{y}} = (0.1, 0.95, 0.05)$:

$$\mathcal{L} = -\log(0.95) = 0.051 \quad \text{(much lower} - \text{better prediction)}$$

Interpretation: The loss only depends on the **predicted probability for the correct class**.

Softmax + Cross-Entropy: Gradient (Step 1)

Let $\hat{y}_k = \text{softmax}(z_k) = \frac{e^{z_k}}{\sum_j e^{z_j}}$ and $\mathcal{L} = -\sum_k y_k \log \hat{y}_k$.

We want: $\frac{\partial \mathcal{L}}{\partial z_i}$ for each logit z_i. First, compute the **Jacobian of softmax**:

Case $i = k$:

$$\frac{\partial \hat{y}_k}{\partial z_k} = \frac{e^{z_k} \sum_j e^{z_j} - e^{z_k} \cdot e^{z_k}}{(\sum_j e^{z_j})^2} = \hat{y}_k - \hat{y}_k^2 = \hat{y}_k(1 - \hat{y}_k)$$

Case $i \neq k$:

$$\frac{\partial \hat{y}_k}{\partial z_i} = \frac{0 - e^{z_k} \cdot e^{z_i}}{(\sum_j e^{z_j})^2} = -\hat{y}_k \hat{y}_i$$

Compact form: $\frac{\partial \hat{y}_k}{\partial z_i} = \hat{y}_k(\delta_{ik} - \hat{y}_i)$ where δ_{ik} is the Kronecker delta.

Softmax + Cross-Entropy: Gradient (Step 2)

Step 2: Apply chain rule with $\mathcal{L} = -\sum_k y_k \log \hat{y}_k$:

$$\frac{\partial \mathcal{L}}{\partial z_i} = \sum_{k=1}^{K} \frac{\partial \mathcal{L}}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_i} = -\sum_{k=1}^{K} \frac{y_k}{\hat{y}_k} \cdot \hat{y}_k(\delta_{ik} - \hat{y}_i)$$

Simplify:

$$= -\sum_k y_k(\delta_{ik} - \hat{y}_i) = -y_i + \hat{y}_i \underbrace{\sum_k y_k}_{=1} = \hat{y}_i - y_i$$

Key Result

$$\boxed{\frac{\partial \mathcal{L}}{\partial z_i} = \hat{y}_i - y_i \qquad \text{(prediction minus truth)}}$$

Same result as BCE + sigmoid! This elegant cancellation is by design.

The Chain Rule

The Chain Rule: Single Variable

If $y = f(u)$ and $u = g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Example: Let $y = (3x + 2)^2$.

- Let $u = 3x + 2$, so $y = u^2$.
- $\frac{dy}{du} = 2u = 2(3x + 2)$.
- $\frac{du}{dx} = 3$.
- $\frac{dy}{dx} = 2(3x + 2) \cdot 3 = 6(3x + 2)$.

Longer chains: If $y = f(g(h(x)))$:

$$\frac{dy}{dx} = \frac{dy}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dx}$$

This is exactly what happens in a neural network: each layer is a function composition.

The Chain Rule: Multiple Variables

If $\mathcal{L} = f(y_1, y_2)$ where $y_1 = g_1(x)$ and $y_2 = g_2(x)$:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y_1}\frac{\partial y_1}{\partial x} + \frac{\partial \mathcal{L}}{\partial y_2}\frac{\partial y_2}{\partial x}$$

In general, if $\mathcal{L}$ depends on x through $y_1, y_2, \ldots, y_m$:

$$\frac{\partial \mathcal{L}}{\partial x} = \sum_{j=1}^{m} \frac{\partial \mathcal{L}}{\partial y_j}\frac{\partial y_j}{\partial x}$$

Example (relevant to neural networks):

A weight $w_{13}^{(1)}$ in layer 1 affects multiple hidden neurons, which all affect the loss:

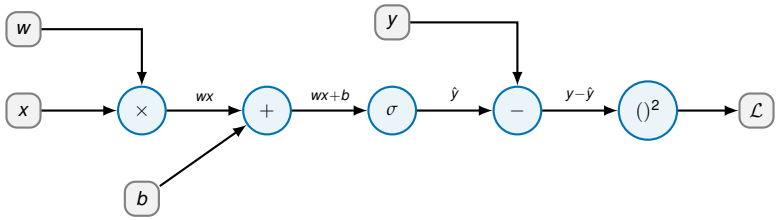
$$\frac{\partial \mathcal{L}}{\partial w_{13}^{(1)}} = \sum_j \frac{\partial \mathcal{L}}{\partial h_j^{(2)}} \cdot \frac{\partial h_j^{(2)}}{\partial w_{13}^{(1)}}$$

We need to sum contributions from all paths through the network.

Computational Graphs

A **computational graph** represents a function as a sequence of operations.

Example: $\mathcal{L} = (y - \sigma(wx + b))^2$



- **Forward pass:** compute output left to right.
- **Backward pass:** compute gradients right to left using chain rule.

Backward Pass Through the Graph

Starting from $\mathcal{L} = (y - \hat{y})^2$, let $e = y - \hat{y}$. Work right-to-left:

$$\frac{\partial \mathcal{L}}{\partial e} = 2e = 2(y - \hat{y}) \qquad \text{(derivative of square)}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\partial \mathcal{L}}{\partial e} \cdot \frac{\partial e}{\partial \hat{y}} = 2(y - \hat{y}) \cdot (-1) = -2(y - \hat{y}) \qquad \text{(chain rule)}$$

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \sigma'(z) = -2(y - \hat{y}) \cdot \sigma'(z) \qquad \text{(through activation)}$$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial z} \cdot x = -2(y - \hat{y})\, \sigma'(z)\, x \qquad \text{(through multiplication)}$$

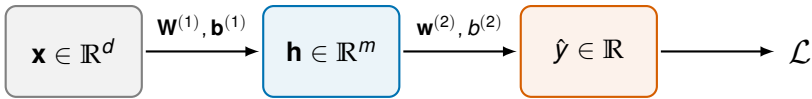
Key pattern: each node passes its gradient backward, multiplied by the local derivative.

Backpropagation

The engine of deep learning

Backpropagation: Setup

Consider a **2-layer network** (1 hidden layer):



Forward pass equations:

$$\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \qquad \mathbf{z}^{(1)} \in \mathbb{R}^m \quad \text{(pre-activation)}$$

$$\mathbf{h} = \sigma(\mathbf{z}^{(1)}) \qquad \mathbf{h} \in \mathbb{R}^m \quad \text{(activation)}$$

$$z^{(2)} = (\mathbf{w}^{(2)})^\top \mathbf{h} + b^{(2)} \qquad z^{(2)} \in \mathbb{R} \quad \text{(output logit)}$$

$$\hat{y} = z^{(2)} \qquad \text{(for regression, no final activation)}$$

$$\mathcal{L} = (y - \hat{y})^2 \qquad \text{(MSE loss)}$$

Goal: Compute $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}}, \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(1)}}, \frac{\partial \mathcal{L}}{\partial \mathbf{w}^{(2)}}, \frac{\partial \mathcal{L}}{\partial b^{(2)}}$.

Backpropagation Step 1: Output Layer Gradients

Start from the loss and work backward.

Step 1a: Gradient of loss w.r.t. output:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}}(y - \hat{y})^2 = -2(y - \hat{y})$$

Since $\hat{y} = z^{(2)}$: $\frac{\partial \mathcal{L}}{\partial z^{(2)}} = -2(y - \hat{y})$. Define $\delta^{(2)} \equiv \frac{\partial \mathcal{L}}{\partial z^{(2)}} = -2(y - \hat{y})$.

Step 1b: Gradient w.r.t. output layer parameters:

$$z^{(2)} = \sum_{j=1}^{m} w_j^{(2)} h_j + b^{(2)}$$

$$\frac{\partial \mathcal{L}}{\partial w_j^{(2)}} = \delta^{(2)} \cdot \frac{\partial z^{(2)}}{\partial w_j^{(2)}} = \delta^{(2)} \cdot h_j \qquad \Rightarrow \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{w}^{(2)}} = \delta^{(2)} \cdot \mathbf{h}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(2)}} = \delta^{(2)}$$

Backpropagation Step 2: Hidden Layer Gradients

Step 2a: Propagate error back to hidden layer. Using chain rule:

$$\frac{\partial \mathcal{L}}{\partial h_j} = \delta^{(2)} \cdot \frac{\partial z^{(2)}}{\partial h_j} = \delta^{(2)} \cdot w_j^{(2)}$$

Step 2b: Pass through the activation function:

$$\frac{\partial \mathcal{L}}{\partial z_j^{(1)}} = \frac{\partial \mathcal{L}}{\partial h_j} \cdot \sigma'(z_j^{(1)}) = \delta^{(2)} \cdot w_j^{(2)} \cdot \sigma'(z_j^{(1)}) \equiv \delta_j^{(1)}$$

Step 2c: Gradients w.r.t. hidden layer parameters (since $z_j^{(1)} = \sum_k W_{jk}^{(1)} x_k + b_j^{(1)}$):

$$\frac{\partial \mathcal{L}}{\partial W_{jk}^{(1)}} = \delta_j^{(1)} \cdot x_k \quad \Rightarrow \quad \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{\delta}^{(1)} \mathbf{x}^\top \qquad \frac{\partial \mathcal{L}}{\partial b_j^{(1)}} = \delta_j^{(1)}$$

The General Backpropagation Formula

Define the **error signal** at layer l: $\boldsymbol{\delta}^{(l)} \equiv \frac{\partial \mathcal{L}}{\partial \mathbf{z}^{(l)}}$

Backpropagation Equations

Output layer ($l = L$): $\boldsymbol{\delta}^{(L)} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} \odot \sigma'(\mathbf{z}^{(L)})$

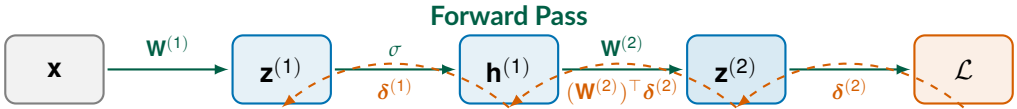
Hidden layers ($l = L-1, \ldots, 1$): $\boldsymbol{\delta}^{(l)} = \left[(\mathbf{W}^{(l+1)})^\top \boldsymbol{\delta}^{(l+1)}\right] \odot \sigma'(\mathbf{z}^{(l)})$

Parameter gradients: $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} (\mathbf{h}^{(l-1)})^\top \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} = \boldsymbol{\delta}^{(l)}$

Here $\odot$ denotes element-wise (Hadamard) product and $\mathbf{h}^{(0)} = \mathbf{x}$.

Intuition: The error at layer l is the error from layer $l+1$ projected back through $\mathbf{W}^{(l+1)}$, scaled by how sensitive the activation was (σ').

Backpropagation: Visual Summary



Algorithm:

1. **Forward:** Compute all $\mathbf{z}^{(l)}$, $\mathbf{h}^{(l)}$, and $\mathcal{L}$ (save intermediate values).
2. **Backward:** Compute $\boldsymbol{\delta}^{(L)}, \boldsymbol{\delta}^{(L-1)}, \ldots, \boldsymbol{\delta}^{(1)}$ using the recursion.
3. **Update:** $\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \alpha\, \boldsymbol{\delta}^{(l)} (\mathbf{h}^{(l-1)})^\top$ for all l.

Backpropagation: Numerical Example (Part 1)

Network: 1 input $\to$ 1 hidden (sigmoid) $\to$ 1 output (linear). MSE loss.

Parameters: $w^{(1)} = 0.5$, $b^{(1)} = 0.1$, $w^{(2)} = -0.3$, $b^{(2)} = 0.2$
Data: $x = 2.0$, $y = 1.0$. **Learning rate:** $\alpha = 0.1$.

Forward pass:

$$z^{(1)} = w^{(1)}x + b^{(1)} = 0.5 \cdot 2.0 + 0.1 = 1.1$$

$$h = \sigma(z^{(1)}) = \sigma(1.1) = \frac{1}{1 + e^{-1.1}} = 0.7503$$

$$z^{(2)} = w^{(2)}h + b^{(2)} = -0.3 \cdot 0.7503 + 0.2 = -0.0251$$

$$\hat{y} = z^{(2)} = -0.0251$$

$$\mathcal{L} = (y - \hat{y})^2 = (1.0 - (-0.0251))^2 = (1.0251)^2 = 1.0508$$

Backpropagation: Numerical Example (Part 2)

Backward pass (work right-to-left through the network):

Step 1 – Error at output:

$$\delta^{(2)} = -2(y - \hat{y}) = -2(1.0 - (-0.0251)) = -2.0502$$

Step 2 – Output layer gradients:

$$\frac{\partial \mathcal{L}}{\partial w^{(2)}} = \delta^{(2)} \cdot h = (-2.0502)(0.7503) = -1.5383 \qquad \frac{\partial \mathcal{L}}{\partial b^{(2)}} = \delta^{(2)} = -2.0502$$

Step 3 – Error at hidden layer:

$$\delta^{(1)} = \delta^{(2)} \cdot w^{(2)} \cdot \sigma'(z^{(1)}) = (-2.0502)(-0.3)(0.7503)(1 - 0.7503) = 0.1154$$

Step 4 – Hidden layer gradients:

$$\frac{\partial \mathcal{L}}{\partial w^{(1)}} = \delta^{(1)} \cdot x = (0.1154)(2.0) = 0.2307 \qquad \frac{\partial \mathcal{L}}{\partial b^{(1)}} = \delta^{(1)} = 0.1154$$

Backpropagation: Numerical Example (Part 3)

Parameter update with learning rate $\alpha = 0.1$: $\theta_{\text{new}} = \theta_{\text{old}} - \alpha \cdot \text{grad}$

Parameter	Old	Gradient	New
$w^{(2)}$	-0.3000	-1.5383	-0.1462
$b^{(2)}$	0.2000	-2.0502	0.4050
$w^{(1)}$	0.5000	0.2307	0.4769
$b^{(1)}$	0.1000	0.1154	0.0885

Verify: Forward pass with new params $\rightarrow \hat{y}_{\text{new}} = 0.2969$

$$\mathcal{L}_{\text{new}} = (1.0 - 0.2969)^2 = 0.4946 \quad \text{(down from 1.0508! ✓)}$$

One gradient step reduced the loss by 53%. Repeat this process thousands of times to train the network.

Vector and Matrix Calculus Notation

In practice, we work with **matrices**, not individual weights.

Key identities (for layer l, with $\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}$):

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} (\mathbf{h}^{(l-1)})^\top \qquad \in \mathbb{R}^{m^{(l)} \times m^{(l-1)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} = \boldsymbol{\delta}^{(l)} \qquad \in \mathbb{R}^{m^{(l)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(l-1)}} = (\mathbf{W}^{(l)})^\top \boldsymbol{\delta}^{(l)} \qquad \in \mathbb{R}^{m^{(l-1)}}$$

Why outer product? Each $\delta_j^{(l)}$ pairs with each $h_k^{(l-1)}$: $\frac{\partial \mathcal{L}}{\partial W_{jk}^{(l)}} = \delta_j^{(l)} \cdot h_k^{(l-1)}$.

The gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}}$ has the same shape as $\mathbf{W}^{(l)}$.

Computational Cost of Backpropagation

- **Forward pass cost:** $O\left(\sum_{l=1}^{L} m^{(l)} m^{(l-1)}\right)$ (matrix-vector products).
- **Backward pass cost:** same order as forward pass.
- **Memory cost:** must store all $\mathbf{z}^{(l)}, \mathbf{h}^{(l)}$ during forward pass.
- **Total:** backprop is roughly **2x** the cost of the forward pass.

Historical Significance

Before backprop (1986): computing gradients for a network with P parameters required $O(P)$ separate forward passes.
With backprop: all gradients in one backward pass.

Gradient Descent Variants

Batch Gradient Descent

Full batch gradient descent uses **all** N samples to compute the gradient:

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \cdot \frac{1}{N} \sum_{i=1}^{N} \nabla_{\mathbf{w}} \ell(\mathbf{x}_i, y_i; \mathbf{w})$$

Pros:

- Stable convergence, low noise.
- Exact gradient direction.

Cons:

- Requires loading **entire dataset** into memory.
- Very slow for large N (one update per epoch).
- Can get stuck in local minima (smooth trajectory).

Not practical when N is large (e.g., ImageNet: $N = 1.2$ million).

Stochastic Gradient Descent (SGD)

SGD uses a **single random sample** per update:

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \cdot \nabla_{\mathbf{w}} \ell(\mathbf{x}_i, y_i; \mathbf{w}) \qquad \text{where } i \text{ is sampled randomly}$$

Pros:

- Very fast updates.
- Can escape shallow local minima (noise helps!).
- Low memory requirement.

Cons:

- Very noisy gradient estimate.
- Oscillates around minimum.
- May never converge exactly.

Key insight: SGD's gradient is an **unbiased estimate** of the true gradient:

$$\mathbb{E}_i \left[\nabla_{\mathbf{w}} \ell(\mathbf{x}_i, y_i; \mathbf{w}) \right] = \frac{1}{N} \sum_{i=1}^{N} \nabla_{\mathbf{w}} \ell(\mathbf{x}_i, y_i; \mathbf{w}) = \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w})$$

Mini-Batch SGD: The Best of Both Worlds

Mini-batch SGD uses a random subset $\mathcal{B} \subset \{1, \ldots, N\}$ of size B:

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \cdot \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla_{\mathbf{w}} \ell(\mathbf{x}_i, y_i; \mathbf{w})$$

Method	Batch size	Noise	Speed
Batch GD	$B = N$	None	Slow
SGD	$B = 1$	High	Fast
Mini-batch SGD	$B = 32, 64, 128, 256$	Moderate	Fast

Why mini-batches work so well:

- GPUs handle parallel matmul — 32 samples $\approx$ as fast as 1.
- Noise reduced by factor $\frac{1}{\sqrt{B}}$ vs. pure SGD.
- Typical: $B = 32, 64, 128, 256$ (powers of 2 for GPU efficiency).

Epochs, Iterations, and Batch Size

Terminology

- **Epoch:** one complete pass through the entire dataset.
- **Iteration:** one parameter update (= one mini-batch).
- **Iterations per epoch:** $\lceil N/B \rceil$.

Example: $N = 10{,}000$ training samples, batch size $B = 100$:

- Iterations per epoch: $10{,}000/100 = 100$.
- If we train for 50 epochs: $50 \times 100 = 5{,}000$ total parameter updates.

Training recipe:

1. Shuffle the dataset.
2. Split into mini-batches of size B.
3. For each mini-batch: forward $\rightarrow$ loss $\rightarrow$ backward $\rightarrow$ update.
4. Repeat for multiple epochs.

SGD with Momentum

Pure SGD oscillates due to noisy gradients. **Momentum** smooths the trajectory:

Momentum Update

$$\mathbf{v}_t = \beta \mathbf{v}_{t-1} + \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}_t) \qquad \mathbf{w}_{t+1} = \mathbf{w}_t - \alpha \mathbf{v}_t$$

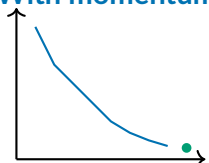
where $\beta \in [0, 1)$ is the momentum coefficient (typically $\beta = 0.9$).

Intuition: a ball rolling down a hill. Momentum accumulates past gradients, so it "rolls through" small bumps.

Without momentum

Oscillates, slow convergence.

With momentum



Smoother, faster convergence.

PyTorch Implementation

Backpropagation in PyTorch: Under the Hood

PyTorch does backpropagation automatically via `autograd`.

```python
import torch
import torch.nn as nn

# 2-layer network: 3 inputs -> 4 hidden (ReLU) -> 1 output
model = nn.Sequential(
    nn.Linear(3, 4),      # W1: 4x3, b1: 4 -> 16 params
    nn.ReLU(),
    nn.Linear(4, 1))      # W2: 1x4, b2: 1 -> 5 params (Total: 21)

# Forward pass + loss
x = torch.randn(1, 3); y_true = torch.tensor([[1.0]])
loss = nn.MSELoss()(model(x), y_true)

# Backward pass -- ALL gradients computed automatically!
loss.backward()

for name, param in model.named_parameters():
    print(f"{name}: grad shape = {param.grad.shape}")
```

Full Training Example: Economic Data

```python
import torch
import torch.nn as nn
from torch.utils.data import DataLoader, TensorDataset

# Simulated economic data: 3 features -> outcome
X = torch.randn(1000, 3)
y = (2*X[:,0] - X[:,1] + 0.5*X[:,2] + torch.randn(1000)*0.1).unsqueeze(1)

# Mini-batch DataLoader
loader = DataLoader(TensorDataset(X, y), batch_size=32, shuffle=True)

# Model, loss, optimizer
model = nn.Sequential(nn.Linear(3, 16), nn.ReLU(),
                      nn.Linear(16, 8), nn.ReLU(), nn.Linear(8, 1))
criterion = nn.MSELoss()
optimizer = torch.optim.SGD(model.parameters(), lr=0.01, momentum=0.9)

# Training loop
for epoch in range(50):
    for X_batch, y_batch in loader:
        loss = criterion(model(X_batch), y_batch)
        optimizer.zero_grad(); loss.backward(); optimizer.step()
```

Visualizing the Training Loss

```
import matplotlib.pyplot as plt

losses = []
for epoch in range(100):
    epoch_loss = 0
    for X_batch, y_batch in loader:
        loss = criterion(model(X_batch), y_batch)
        optimizer.zero_grad(); loss.backward(); optimizer.step()
        epoch_loss += loss.item()
    losses.append(epoch_loss / len(loader))

plt.plot(losses); plt.xlabel('Epoch'); plt.ylabel('MSE Loss')
plt.title('Training Loss Curve'); plt.grid(True); plt.show()
```

What to look for: loss should decrease; oscillations $\Rightarrow$ LR too large; slow decrease $\Rightarrow$ LR too small.

Summary & Next Steps

Lecture 2 Summary

1. **Activation function derivatives** are critical for backprop. Sigmoid: $\sigma' = \sigma(1-\sigma)$. ReLU: $\sigma' = \mathbb{1}[z > 0]$.
2. **Vanishing gradients:** sigmoid derivative ≤ 0.25 causes exponential decay in deep nets. ReLU solves this.
3. **Softmax** extends binary classification to K classes: $\hat{y}_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$.
4. **Cross-entropy + softmax/sigmoid** gives gradient $\frac{\partial \mathcal{L}}{\partial z_i} = \hat{y}_i - y_i$.
5. **Backpropagation** = chain rule applied layer by layer:

$$\boldsymbol{\delta}^{(l)} = [(\mathbf{W}^{(l+1)})^\top \boldsymbol{\delta}^{(l+1)}] \odot \sigma'(\mathbf{z}^{(l)}) \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} (\mathbf{h}^{(l-1)})^\top$$

6. **Mini-batch SGD** with momentum is the standard optimizer. Typical $B = 32$–256.

Next Lecture: SGD, Regularization & Generalization

Lecture 3 (Monday, April 20):

- Advanced optimizers: Adam, AdaGrad, RMSProp.
- Regularization: L1, L2 (weight decay), dropout.
- Batch normalization.
- Universal Approximation Theorem; why depth matters.
- Practical tips: LR schedules, early stopping.

Recommended reading: Goodfellow et al., *Deep Learning*, Ch. 6.5 + Ch. 8. Zhang et al., *D2L*, Ch. 5 (`https://d2l.ai`).

Homework 1 will be released after Lecture 3 (due Sunday, April 27).

Questions?

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